

Merge Sort (Recursive)

DAA Practical 8 — 5th Sem CSE B.Tech

Source Code

```
// Aim: implement Merge Sort using recursion
#include <stdio.h>

void merge(int arr[], int l, int m, int r) {
    int n1 = m - l + 1, n2 = r - m;
    int left[50], right[50];
    for (int i = 0; i < n1; i++) left[i] = arr[l + i];
    for (int j = 0; j < n2; j++) right[j] = arr[m + 1 + j];

    int i = 0, j = 0, k = l;
    while (i < n1 && j < n2)
        arr[k++] = (left[i] <= right[j]) ? left[i++] : right[j++];
    while (i < n1) arr[k++] = left[i++];
    while (j < n2) arr[k++] = right[j++];
}

void mergeSort(int arr[], int l, int r) {
    if (l < r) {
        int m = (l + r) / 2;
        mergeSort(arr, l, m);
        mergeSort(arr, m + 1, r);
        merge(arr, l, m, r);
    }
}

int main() {
    int arr[] = {38, 27, 43, 3, 9, 82, 10};
    int n = 7;

    printf("Original array: ");
    for (int i = 0; i < n; i++) printf("%d ", arr[i]);
    printf("\n");

    mergeSort(arr, 0, n - 1);

    printf("Sorted array:  ");
    for (int i = 0; i < n; i++) printf("%d ", arr[i]);
    printf("\n");

    return 0;
}
```

Output

```
Original array: 38 27 43 3 9 82 10
Sorted array:   3 9 10 27 38 43 82
```